

MACHINEPULSE · DESIGN DECISION RECORD

MachinePulse recommendations accepted

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DOCUMENT CONTROL

Document ID	MPL-DDR-002	Version	0.1
Status	Draft	Date	2026-09-25
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/machinepulse		
Source	docs/decisions/0002-recommendations-accepted.md @ 2616d17		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-09-25	Amish Chadha	Record the recommendations accepted by Amish on 2026-09-25, what changed in the repo, and the items still open

0002: Recommendations accepted

- **Date:** 2026-09-25
- **Status:** accepted. Amish accepted every recommendation in this repo on 2026-09-25. Items with no recommendation stay "Proposed, awaiting Amish".

■ Context

On 2026-09-25 Amish wrote: "i accept all your recommendations, go with them across all repos." Before that, MachinePulse held ten items adopted for TRL 3 work and open for his review (MPL-DDR-001, D1 to D10), three new proposals from the TRL 3 calculations (docs/REVIEW.md , session "TRL 3", items 3 to 5) and two items without a single recommendation (MPL-DDR-001, O1 and O2).

■ Options considered

The options for each item are those in docs/REVIEW.md (sessions "/populate" and "TRL 3") and in MPL-DDR-001. Where a recommendation named one of several options, that option is the decision.

■ Decision

Table 1. Items decided by Amish, 2026-09-25: go with recommendation.

#	Item	Decision	What changed in the repo
D1	Radio	Wi-Fi on the ESP32-S3 first; LoRaWAN summaries through FieldNode a later option	Status wording only (MPL-DDR-001 v0.2, MPL-PRC-001 v0.4)
D2	Power	Certified 5 V USB adapter	Status wording only
D3	Accelerometer	IIS3DWB class, ADXL345 class as documented fallback	Status wording only
D4	Current sensing	One CT on one phase	Status wording only (MPL-PRB-001 v0.4)
D5	Alert logic	First-week baseline per load band; flags to a person	Status wording only
D6	Data home	Twinkit gateway, any MQTT broker, no cloud	Status wording only
D7	Enclosure	Stock IP54 ABS box on 5 mm nylon stand-offs	Status wording only
D8	Pilot site	A makerspace or university workshop with a lathe and a compressor	Status wording only (MPL-PRB-001 v0.4); choosing the actual site and running the pilot are TRL 4 work, on hold
D9	Budget	No change at the time; superseded in effect by N1	See N1
D10	Data schedule	Features every minute, a spectrum every 15 min	Status wording only
N1	Magnets for hot frames (R12)	Option (b): high-temperature pot magnets rated 120 °C or more, about \$1.00 more for the pair	<code>cad/src/model.py</code> gains <code>mag_t_max = 120 °C</code> (geometry unchanged); STEP and STL re-exported; BOM line 6 from \$3.50 to \$4.00 each, parts from \$80.00 to \$81.00; <code>budget_usd</code> from 80 to 81; GA MPL-DWG-001 from Rev P1 to Rev P2 (magnet note); MPL-CAL-001 v0.2: hot-frame limit from 81.3 °C (magnets) to 109.6 °C (accelerometer), R12 from at risk to met by calculation; R16 target from \$80 to \$81 in MPL-REQ-001 v0.4
N2	R8 range	Restate R8 as within 2 °C from 0 to 85 °C and within 4 °C from 85 to 100 °C	MPL-REQ-001 v0.4; R8 from at risk to met by calculation (1.69 K at 85 °C, 3.59 K at 100 °C)
N3	R3 and R5 follow-up	Keep both targets; measure the ADC residual and the mount resonance first at TRL 4; consider a flat steel saddle for curved frames	Targets unchanged; noted in MPL-CAL-001 v0.2 and MPL-PRC-001 v0.4. The measurements and any saddle design are TRL 4 work, decided but on hold

Table 2. Items still open.

#	Item	Status
01	Co-design partner for the alerts and dashboard. No recommendation was made.	Proposed, awaiting Amish
02	How to treat R4 with one CT: accept at risk with a power factor curve, relax to "relative energy trend", or add a voltage reference (\$10, parts \$91.00). The recommendation offered two courses without choosing one, so no option is taken.	Proposed, awaiting Amish

■ Consequences

- Requirement status (MPL-CAL-001 v0.2): 1 not met (R10), 4 at risk (R3, R4, R5, R16), 9 met by calculation, 3 met by design. Before: 1 not met, 6 at risk, 7 met by calculation, 3 met by design.
- `project.yaml`: `budget_usd` from 80 to 81; `trl` and `trl_target` stay 3. No reworded pitch or problem was recommended, so those lines are unchanged.
- Controlled documents revised: MPL-PRB-001 v0.4, MPL-PRC-001 v0.4, MPL-REQ-001 v0.4, MPL-CAL-001 v0.2, MPL-DDR-001 v0.2; drawing MPL-DWG-001 Rev P2; concept media regenerated.
- Cross-repo actions: none required. The decisions do not change any interface with TwinKit, FieldNode or CalRig.
- TRL 4 is on hold by Amish's instruction. Nothing in this record authorizes building, testing or buying parts.